

Technical Assistance by JICA



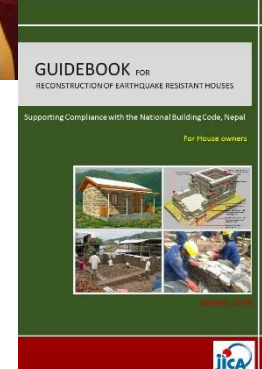
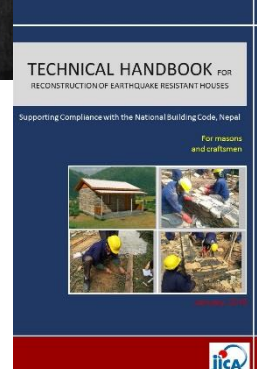
DESIGN CATALOGUE FOR RECONSTRUCTION OF EARTHQUAKE RESISTANT HOUSES

Volume I



Nepal Housing
Reconstruction Programme

Government of Nepal
Ministry of Urban Development
Department of Urban Development and Building Construction
Babarmahal, Kathmandu



TRANSITIONAL PROJECT IMPLEMENTATION SUPPORT
FOR EMERGENCY RECONSTRUCTION PROJECTS IN NEPAL

TPIS-ERP



for Reconstruction of earthquake resistant houses

Field Level

- **Mason Training**
- **House-owner Training**
- **TOT for DUDBC Engineer**
- **Mobile Team for Survey and Assistance/Public Awareness Campaign**
- **Inspection Training for DUDBC Engineers**
- **Refresher Training for Mason**

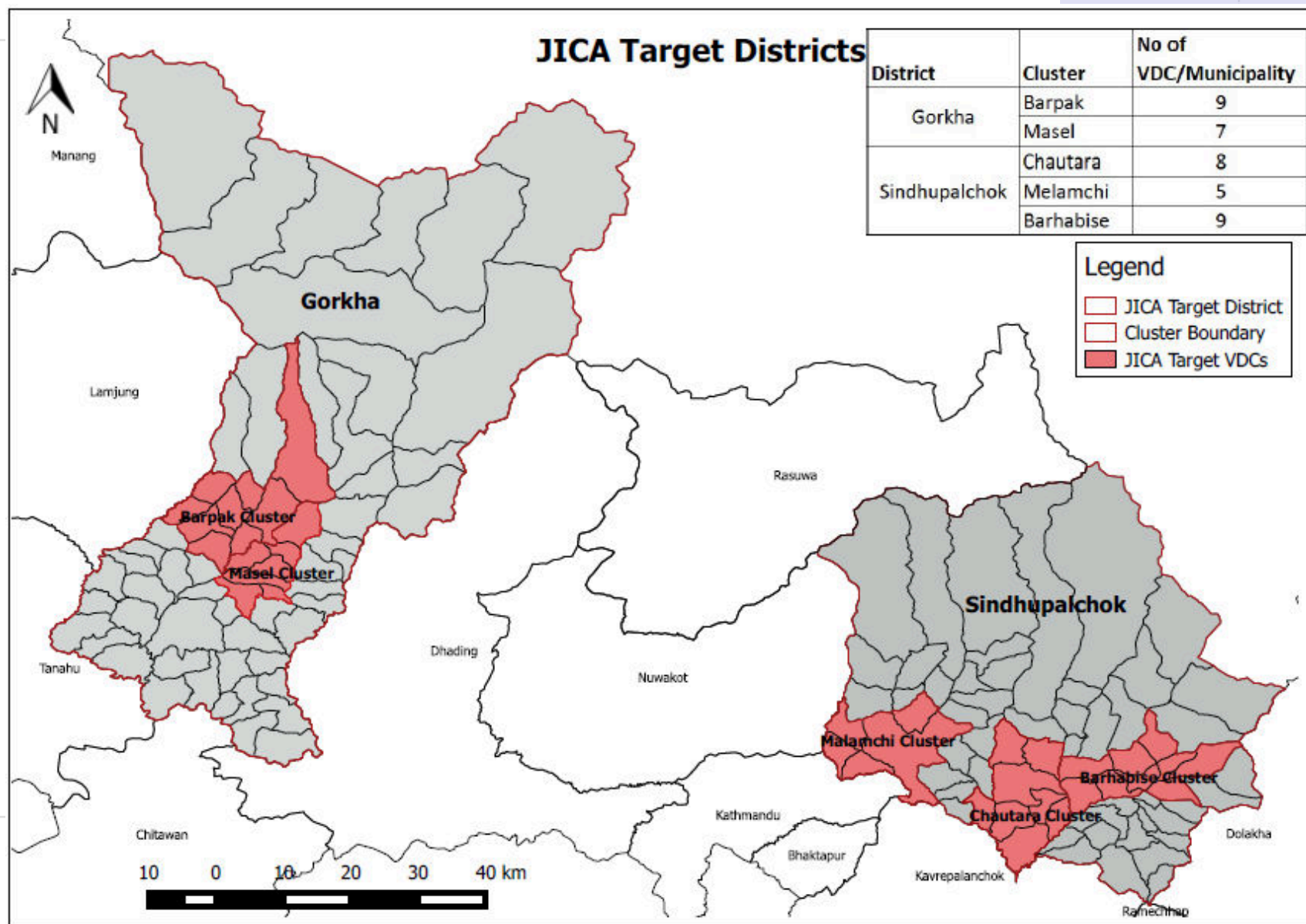
Central Level

- **Housing Design Catalogue**
- **Minimum Requirements for masonry structure, (Poster, Handbook)**
- **Training materials (Guidebook, Handbook, Technical Video)**
- **Inspection Manual, Inspection training curriculum**
- **Correction Manual (A part of Retrofitting manual)**
- **Compliance Catalogue**

JICA Target Districts

TPIS-ERP

for Reconstruction of earthquake resistant houses

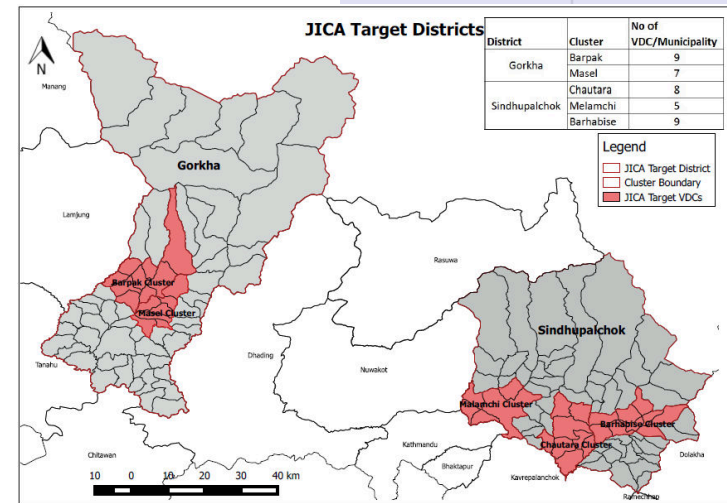


Mason Training

Total Number of Trained mason :

1431

(Gorkha: 427, Sindhupalchok: 1004)

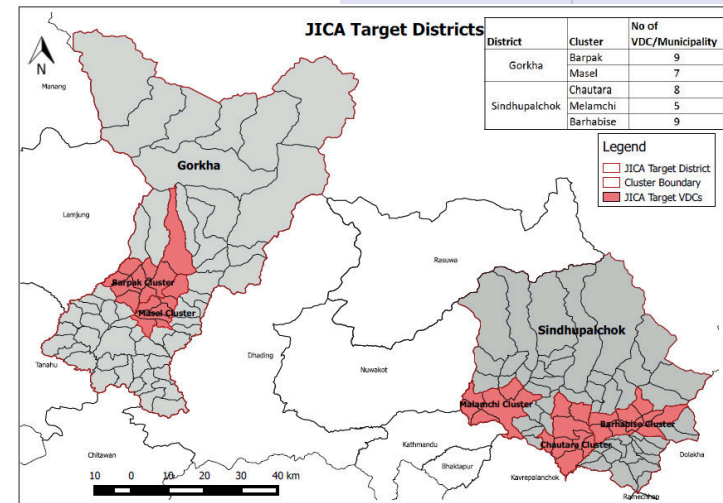


House-owner Training

Total Number of House-owner:

3647

(Gorkha: 1065, Sindhupalchok: 2582)

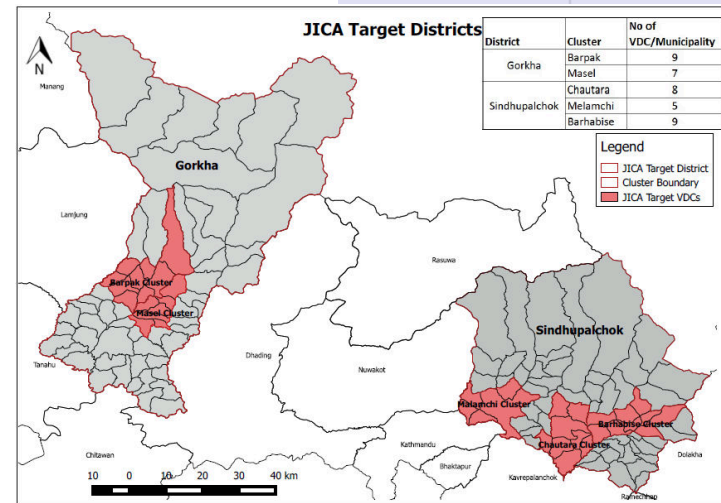


for Reconstruction of earthquake resistant houses

➤ TOT for DUDBC Engineer

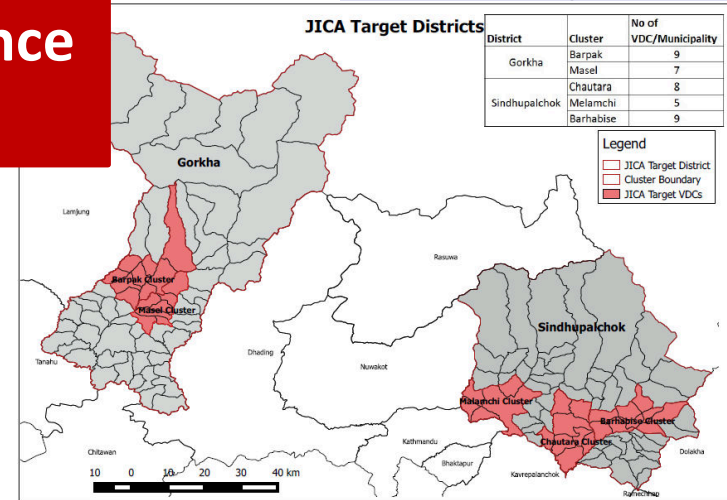
Total Number of Engineer:
440

	Engineer	Sub Engineer	Assistant of Sub Engineer
Gorkha	115	28	46
Sindhupalchok	139	54	58



for Reconstruction of earthquake resistant houses

- Mobile Team for Survey and Assistance
- Public Awareness Campaign



for Reconstruction of earthquake resistant houses

➤ Housing Design Catalogue, volume 1

DESIGN CATALOGUE FOR RECONSTRUCTION OF EARTHQUAKE RESISTANT HOUSES

Volume I



October, 2015 (Aswin, 2072)



Nepal Housing
Reconstruction Programme

Government of Nepal
Ministry of Urban Development
Department of Urban Development and Building Construction
Babarmahal, Kathmandu

Model SMC-2.1: STONE MASONRY IN CEMENT MORTAR

TWO-STOREY



LEVEL	MAN POWER			MATERIALS							
	Skilled	Unskilled	Stone	CEMENT	SAND	AGGREGATE	WOOD	CGI SHEET	Reinforcing bar		
	Md		cum.	Bags	Cum.	Cum.	Cum.	Bundle	Kg		
Up to Plinth Level	88	317	17	76	12	6	0	0	148		
SUPERSTRUCTURE	175	290	31	111	17	4	2.95	1.02	631		
ROOFING	17	20	0	0	0	1.48	3.69	0	0		
TOTAL	279	636	48	187	29	10	4.43	4.71	779		



Nepal Housing
Reconstruction Programme

TYPE OF HOUSE:
MODEL SMC-2.1
DRAWING TITLE:
PERSPECTIVE AND ESTIMATION

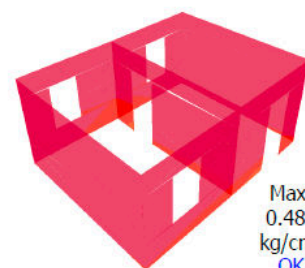
SCALE: None
DESIGNED BY: JICA

DATE:
SMC-2.1
1/4

35cm thick
brick-wall
with

cement
mortar

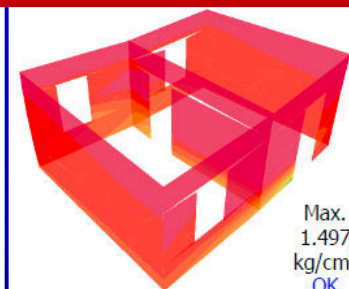
With seismic
bands
(plinth, sill, lintel,
roof band)



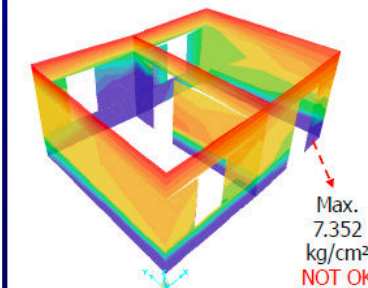
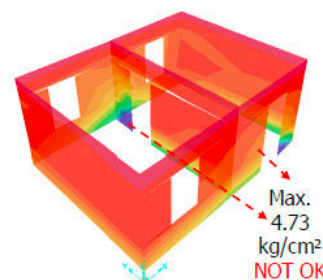
Maximum Brick Tensile Stress

NBC/IS

JMA Kobe



Without seismic
bands



Teddy Boen

for Reconstruction of earthquake resistant houses

➤ Structural calculation/ analysis

	(MPa)	(MPa)	(MPa)	(MPa)																																																							
SMC-1.1:					STONE MA																																																						
	9.165	0.3375	0.332		All tensile stresses are below the limit. OK!																																																						
SMC-1.2:					STONE MA																																																						
	9.165	0.3375	0.847		- Almost all tensile stresses are below the limit. - The stresses are slightly exceeded only at door opening.																																																						
SMC-2.1:					STONE MA																																																						
	9.165	0.3375	0.618		Almost all tensile stresses are below the limit. At Ground Floor, the tensile stresses exceeded at door openings in the front elevation and door between two rooms.																																																						
SMC-2.2:					STONE MA																																																						
	9.165	0.3375	0.780		Almost all tensile stresses are below the limit. At the Ground Floor, the tensile stresses exceeded at door openings in the front veranda and door between two rooms.																																																						
SMC-2.3:					STONE MA																																																						
	9.165	0.3375	7.774		At Ground Floor, the tensile stresses exceeded at door and window openings in the front elevation and spread to																																																						
<table border="1"> <thead> <tr> <th colspan="5">Assumptions</th><th></th></tr> <tr> <th>Model</th><th>E Walls</th><th>Tensile strength</th><th>Max. Tensile</th><th>Wall tensile stress pattern</th><th>Notes for walls</th></tr> <tr> <th></th><th>(MPa)</th><th>(MPa)</th><th>(MPa)</th><th>(MPa)</th><th></th></tr> </thead> <tbody> <tr> <td>SMC-2.4:</td><td></td><td></td><td></td><td></td><td>STONE MA</td></tr> <tr> <td></td><td>9.165</td><td>0.3375</td><td>2.906</td><td></td><td>At Ground Floor, the tensile stresses exceeded at door and window openings in the front elevation and spread to perpendicular walls.</td></tr> <tr> <td>SMC-2.5:</td><td></td><td></td><td></td><td></td><td>STONE MASONR</td></tr> <tr> <td></td><td>9.165</td><td>0.3375</td><td>2.97</td><td></td><td>- At Ground Floor, the tensile stresses exceeded at door and window openings in the front elevation and spread to perpendicular walls. - At 1st Floor, almost all tensile stresses are below the limit; the tensile stresses are only slightly exceeded at door openings in front elevation.</td></tr> <tr> <td>SMC-2.6:</td><td></td><td></td><td></td><td></td><td>STONE MASONR</td></tr> <tr> <td></td><td>9.165</td><td>0.3375</td><td>2.349</td><td></td><td>At the Ground Floor, the tensile stresses are exceeded at the window openings.</td></tr> </tbody> </table>						Assumptions						Model	E Walls	Tensile strength	Max. Tensile	Wall tensile stress pattern	Notes for walls		(MPa)	(MPa)	(MPa)	(MPa)		SMC-2.4:					STONE MA		9.165	0.3375	2.906		At Ground Floor, the tensile stresses exceeded at door and window openings in the front elevation and spread to perpendicular walls.	SMC-2.5:					STONE MASONR		9.165	0.3375	2.97		- At Ground Floor, the tensile stresses exceeded at door and window openings in the front elevation and spread to perpendicular walls. - At 1st Floor, almost all tensile stresses are below the limit; the tensile stresses are only slightly exceeded at door openings in front elevation.	SMC-2.6:					STONE MASONR		9.165	0.3375	2.349		At the Ground Floor, the tensile stresses are exceeded at the window openings.
Assumptions																																																											
Model	E Walls	Tensile strength	Max. Tensile	Wall tensile stress pattern	Notes for walls																																																						
	(MPa)	(MPa)	(MPa)	(MPa)																																																							
SMC-2.4:					STONE MA																																																						
	9.165	0.3375	2.906		At Ground Floor, the tensile stresses exceeded at door and window openings in the front elevation and spread to perpendicular walls.																																																						
SMC-2.5:					STONE MASONR																																																						
	9.165	0.3375	2.97		- At Ground Floor, the tensile stresses exceeded at door and window openings in the front elevation and spread to perpendicular walls. - At 1st Floor, almost all tensile stresses are below the limit; the tensile stresses are only slightly exceeded at door openings in front elevation.																																																						
SMC-2.6:					STONE MASONR																																																						
	9.165	0.3375	2.349		At the Ground Floor, the tensile stresses are exceeded at the window openings.																																																						

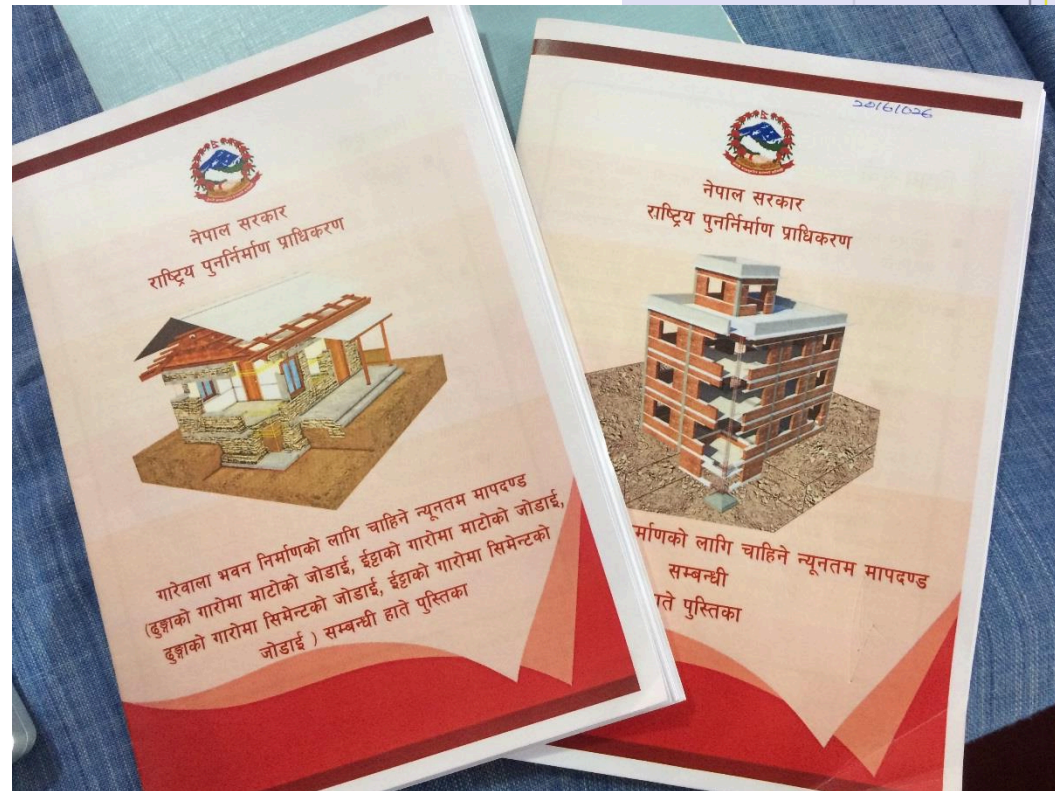
BMC-1.1:					BRICK MA
	258	0.167	0.076		All tensile stresses are below the limit. OK!
BMC-1.2:					BRICK MA
	258	0.167	0.106		All tensile stresses are below the limit. OK!
BMC-2.1:					BRICK MA
	258	0.167	0.125		All tensile stresses are below the limit. OK!
BMC-2.2:					BRICK MA
	258	0.167	0.547		At ground level, the tensile stress in the door/window openings are exceeded.
BMC-2.3:					BRICK MA
	258	0.167	0.401		- Almost all tensile stresses are below the limit. - At Ground Floor, the tensile stresses are exceeded at window opening in left elevation.
BMC-2.4:					BRICK MASONR
	258	0.167	0.431		- Almost all tensile stresses are below the limit. - At 1st Floor, the tensile stresses are exceeded at window openings in front and left elevation.
BMC-2.5:					BRICK MASONR
	258	0.167	0.209 (coupled 0.194)		- Almost all tensile stresses are below the limit. - At Ground Floor, the tensile stresses are slightly exceeded at door openings

SMM-1.1:					STONE M
	300	0.056	using wooden bands: 0.222 using RC bands: 0.101		- For walls using wooden horizontal bands, tensile stresses at door openings are exceeded. - For walls using RC horizontal bands, tensile stresses at door openings are slightly exceeded. However, with RC bands, the tensile stresses are reduced.
BMM-1.1:					BRICK M
	170	0.05	using wooden bands: 0.192 using RC bands: 0.036		- For walls using wooden horizontal bands, tensile stresses at door openings are exceeded. - For walls using RC horizontal bands, all tensile stresses are below the limit.

for Reconstruction of earthquake resistant houses

➤ Minimum Requirements

Minimum Requirements (MRs) for Stone Masonry in Cement Mortar (NBC202)			Page1
No.	Category		
1	Site Selection	A building shall not be constructed if site is:	
		✓ Geological fault or Raptured Area	
		✓ Areas Susceptible to Landslide	
		✓ Steep Slope > 20%	
		✓ Filled Area	
2	Shape of House	✓ River Bank and Water-logged Area	
		No. of story	✓ Up to two-storied load-bearing masonry buildings constructed in cement mortar.
		Span of wall	✓ The span of wall shall not more than 4.5 meters
		Size of room	✓ The area of individual floor panel not more than 13.5 square metres
		Height of wall	✓ The height of wall should not be more than 3.0 meters
3	Foundation	Proportion	✓ The house shall be planned in square, rectangular. Avoid long and narrow structure should not be more than 3 times of its width.
		General	✓ The foundation trench shall be of uniform width. The foundation bed shall be on the same level throughout the foundation in flat area.
		Depth	✓ The depth of footing should not be less than 800mm for one story, 900mm for two storey.
		Width	✓ The width of footing should not be less than 600mm in medium soil condition. As depend on soil condition. Shown in detail drawings.
4	Plinth		
		General	✓ Provide a reinforced concrete band at plinth level, as shown in detail drawings. The top level of plinth should not be less than 300mm from existing ground level. Recommendation is 450mm.
		Height	✓ Minimum height of Plinth band is 150mm.
		Width	✓ Minimum thickness of plinth band width should be equal to wall thickness. 350mm for Stone masonry.
		Reinforcement	✓ Main reinforcement should be 4-12 dia. bars. Use 6mm diameter rings at 150mm. Hook length should be 500mm. Bars shall have a clear cover of 25mm concrete.
5	Walls	General	✓ Masonry should not be laid staggered or straggled in order to avoid continuous vertical joints. At corners or wall junctions, through vertical joints should be avoided by properly laying the masonry. It should be interlocked.
		Joints	✓ Mortar joints should not be more than 20mm and less than 10mm in thickness. The ratio recommend 1:4 (Cement: Sand).
		Through Stone	✓ Through-stone of a length equal to the full wall thickness should be used in every 600 mm lift at not more than 1.2 m apart horizontally.
		Width	✓ The minimum width of wall is 350mm for one-storey and two-storey.



Handbook of MR

MR of SMC

for Reconstruction of earthquake resistant houses

➤ Minimum Requirement

ढुङ्गाको गारोमा सिमेन्ट मसलाको जोडाई - निर्माण प्रक्रिया
राष्ट्रिय भवन संहिता २०२ मा आधारित
नेपाल सरकार
राष्ट्रिय कृषिविकास प्राधिकरण

The poster illustrates the construction sequence for a stone masonry house with a concrete frame. It includes diagrams for the foundation, walls, roof, and interior finishing. A 3D perspective view of the completed house is shown in the center. The poster is divided into sections for different parts of the house, such as the foundation, walls, roof, and interior finishing.

Construction Sequence Poster

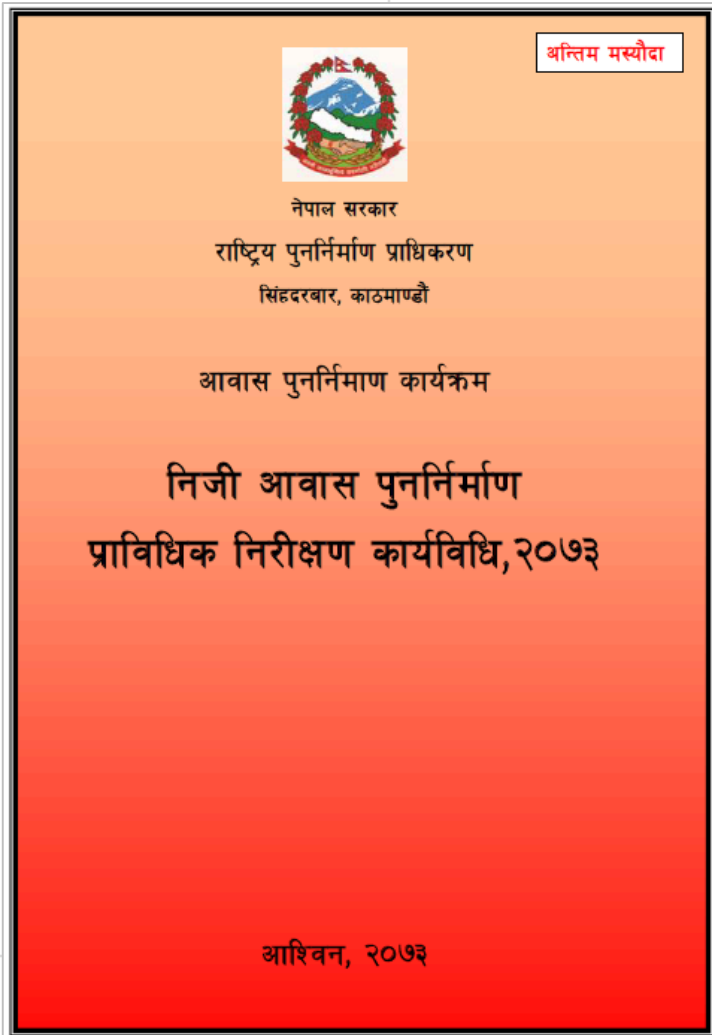
ढुङ्गाको गारोमा सिमेन्ट मसलाको जोडाई—न्यूनतम मापदण्ड
राष्ट्रिय भवन संहिता २०२ मा आधारित
नेपाल सरकार
राष्ट्रिय कृषिविकास प्राधिकरण

The poster illustrates the minimum requirements for building a stone masonry house with a concrete frame. It includes diagrams for the foundation, walls, roof, and interior finishing. A 3D perspective view of the completed house is shown in the center. The poster is divided into sections for different parts of the house, such as the foundation, walls, roof, and interior finishing. It also includes a table for the minimum requirements for different parts of the house.

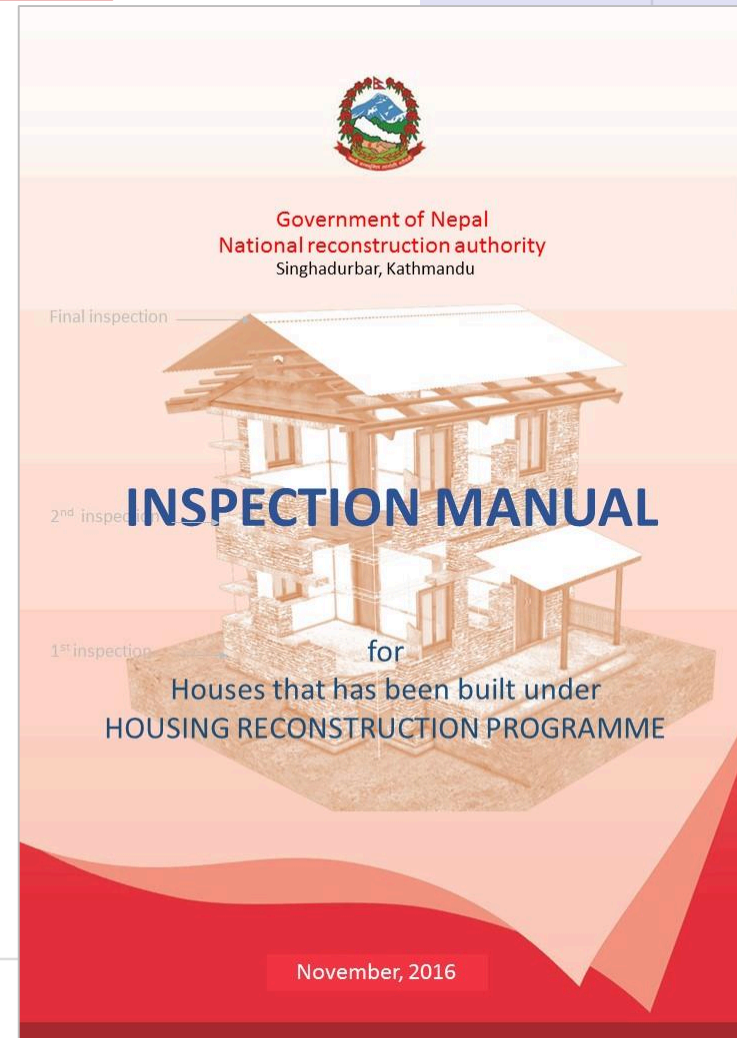
Minimum Requirements Poster

for Reconstruction of earthquake resistant houses

➤ Inspection Materials



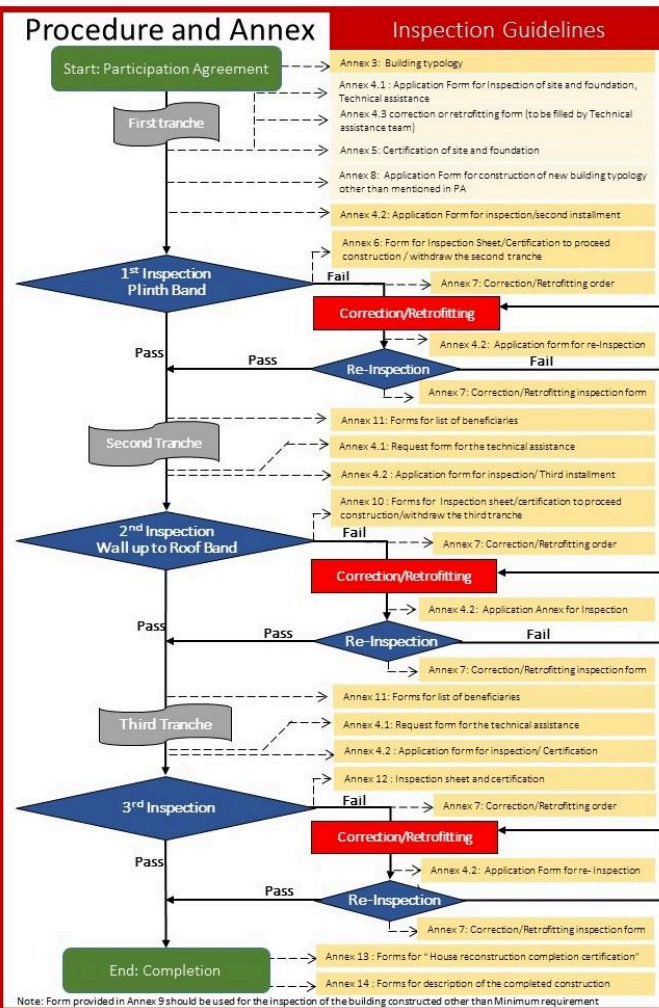
NRA Inspection Guidelines



Inspection Manual

for Reconstruction of earthquake resistant houses

➤ Inspection Guidelines



Annex 3.2.a: Form: Inspection Sheet of Stone Masonry with Mud Mortar for First Inspection

Government of Nepal
Ministry of Urban Development
Central Level Project Implementation Unit

Form No. 3. 2. a : Serial No. Grand Agreement No.

Inspection Sheet of Stone Masonry with Mud Mortar for First Inspection

Information of House Owner/Beneficiary

Name: Address: Date of Inspection: Design No. SVM-1.1

SECTION-I: DESCRIPTION PROVIDED IN THE APPLICATION TO SURVEY THE HOUSE

If free design by house owner: Technical Assistant: YES/NO Organization: CoOp/NGO () Trained Masons used: YES/NO Soil type: Hard/Medium/Soft

SECTION-II: DETAILED TECHNICAL INSPECTION

MR No.	Category	Description	Comply to MRs	Remarks
			YES NO	
1	Site selection located away from	Geological fault or Ruptured Area Steep Slope > 20° Landslide susceptible Area River bank and Water logged Area Rock Fall Area Liquefaction susceptible Area Filled Area	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	
2	Shape of House	No. of storey: RC band: Not more than one plus habitable attic; Timber band: Not more than one storey. Span of Wall: Not more than 12 times thickness of wall and not more than 4.5m (14'9") Size of room: Not more than 13.5sq.m (145'4sq.ft) Proportion: Simple and regular shape as square and rectangular The length is not more than 3 times of its width	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	
3	Materials	Stone: Avoid round, easily breakable soft stone; Size: Thickness 50mm (2") length 300mm (12") Mortar: Free from organic materials, pebbles, hard materials. Concrete: M15 grade (1 cement; 2 sand; 4 aggregate) Rebar: fy = 415 Mpa / 5000 Mpa Timber: Hard wood	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	
4	Foundation	Continuous strip footing Depth below GL: 750mm (2'6") for one storey; Soft soil 800mm (2'8") Medium 750mm (2'6") Hard 750mm (2'6") Base Width: Started right from the foundation	☐ ☐ ☐ ☐ ☐ ☐	
5	Vertical member	Reinforcement: RC band: Placed at all corners, junctions of walls and openings; Timber band: Hard wood, One member of 75mm (3") x 100mm (4") for corner, Two member of 75mm (3") x 100mm (4") for openings. Anchorage: 50 times diameter of reinforcement Height from GL: Not less than 300mm (12") from GL Thickness: 150mm (6") for medium and soft soil, 75mm (3") for hard soil Width: Not less than wall thickness/300mm	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	
6	Plinth	Reinforcement: RC band: Main: 4-12dia for 150mm (6"); 3-10dia for 75mm (3") height; 50mm (2") dia. at 150mm (6") Concrete cover of 25mm (1") Timber band: Main member: 2-75mm (3") connected with 50mm (2") batten	☐ ☐ ☐ ☐	

Annex 3.2.a: Form: Inspection Sheet of Stone Masonry with Mud Mortar for First Inspection

a) After the detail description of the under constructed house, is it satisfactory to give permit for the further construction. Yes ☐, No ☐
If it is satisfactory, then please provide inspection certificate based on Annex-2, if not please order for correction based on Annex-4.

b) If found unsatisfactory, what are the things that need to be improve:

1.
2.
3.
4.

Others:

1. At least four number of photographs with their number

2. Tentative drawings of building:

c) I agreed the above result of inspection conducted by the inspector. Yes ☐, No ☐
House owner/Beneficiaries or representative: Name: Signature: Date: Relationship with house owner (in case of representative):
Inspected by: Name: Signature: Date:
Approved by: Name: Signature: Date:

Note:
• Different forms are available for different building types.
• Use appropriate form

Inspection Sheet

Procedure and Forms

for Reconstruction of earthquake resistant houses

Inspection Training for DUDBC Engineer

Time		Session	Day 1	Session	Day 2
9:00	9:30	S 1.0	Opening Ceremony	S 2.0	Review of Previous day
9:30	10:15	S 1.1	Inspection Guidelines and Forms	S 2.1	Class Exercise for Inspection
10:15	11:00	S 1.2	Responsibility of stake holder		
11:00	11:45	S 1.3	MIS and Use of Tablet	S 2.2	Field Visit Briefing and Preparation
11:45	12:45	Lunch		Lunch	
12:45	13:30	S 1.4	Overview of Construction Sequence	S 2.3	Field Visit
13:30	14:15	S 1.5	General consideration for Construction		
14:15	14:45	Tea Break		Tea Break	
14:45	15:30	S 1.6	Masonry Procedure	S 2.4	Field Visit discussion
15:30	16:15	S 1.7	R.C. frame Procedure		
16:15	17:00	S 1.8	Review of the day	S 2.5	Closing and Certification

Field Level

- Mason Training
- House-owner Training
- TOT for DL-PIU Engineer
- Mobile Team for Survey and Assistance/Public Awareness Campaign
- **Inspection Training for DUDBC Engineers**
- **Refresher Training for Mason**

Central Level

- Housing Design Catalogue, Vol. 1
- Minimum Requirements for masonry structure, (Poster, Handbook)
- Training materials (Guidebook, Handbook, Technical Video)
- Inspection Manual, Inspection training curriculum
- **Housing Design Catalogue, Vol. 2**
- **Collection Manual (A part of Retrofitting)**

Activity Plan from now on

for Reconstruction of earthquake resistant houses

➤ Housing Design Catalogue, Vol. 2

Approach

Design Catalogue
Volume 1

Basic model of
NBC 202, 203

Design Catalogue
Volume 2

New technic/Method
Rammed earth
CSEB, etc

Design Catalogue
Volume 3

for urban model?

Design Catalogue
Volume 4

**NBC105
or IS**

or International
Seismic code

◆ Need same criteria for seismic design

◆ Need Structural calculations

- ◆ There is no specification of Timber structure, Concrete Block, Stabilized Soil Block, etc on NBC.
- ◆ It takes time to set Minimum Requirement of these methods.
- ◆ It is better to make fix design in the catalogue.

Model SMC: STONE MASONRY IN MUD MORTAR

ONE-STOREY

LEVEL	MAN POWER		MATERIALS								
	Skilled	Unskilled	Stone	CEMENT	SAND	AGGREGATE	Mud	WOOD	CGI SHEET	PLAIN SHEET	ROD
	Md	Md	Cu.m	Bags	Cu.m	Cu.m	Cu.m	Cu.m	Bundel	Rm	kg
Up to Plinth Level	44	155	37	6	0.42	0.8	39.0	0	0	0	91.6
Super-structure	62	55	17	15	1.03	2.1	6.36	0.79	0	0	221.3
Roofing work	35	13	0	0	0	0	0	1.48	4.71	8	0
TOTAL	141	221	54	21	1.45	2.9	45.4	2.27	4.71	8	312.9

Nepal Housing
Reconstruction Programme

TYPE OF HOUSE:

MODEL SMC-

DRAWING TITLE:

PERSPECTIVE AND ESTIMATION

SCALE:

None

DATE:

DESIGNED BY:

JICA

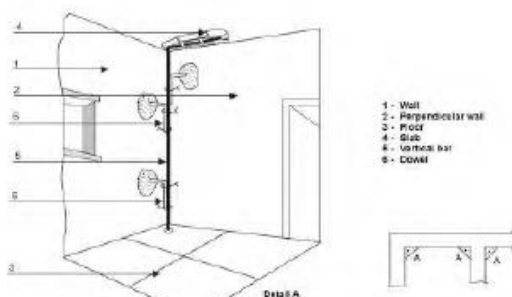
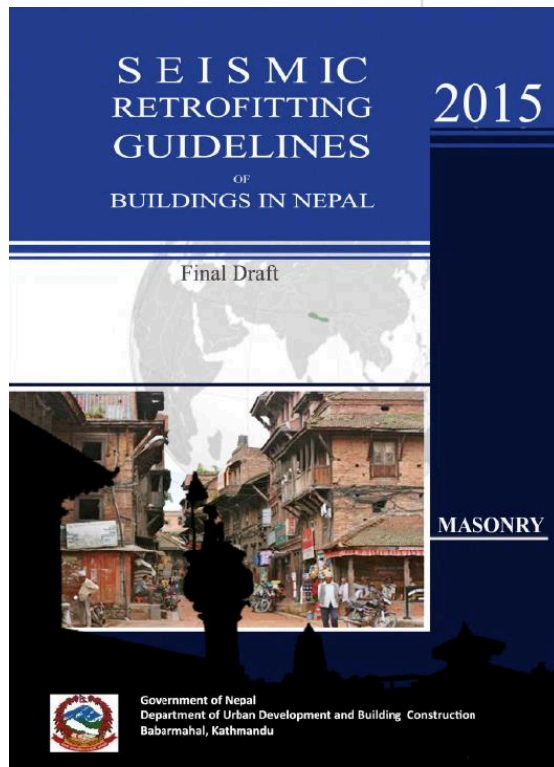
SMC-

1/4

2



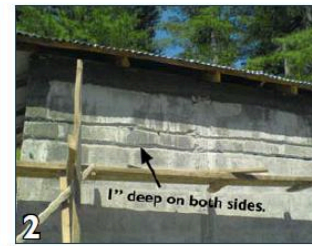
➤ Correction Manual



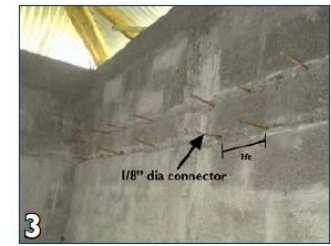
Ex. Adding Vertical element



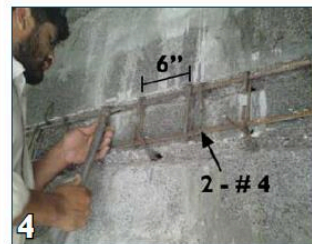
1 Clean the surface and drill holes.



2 Make grooves on both faces to place reinforcement.



3 Insert connectors at every 1 ft.



4 Place reinforcement at both faces and fix them firmly with 1/8" dia connectors.



5 Place shuttering and pour concrete.



6 Cure concrete for at least 10 days.

Ex. Adding horizontal band

Target: New construction

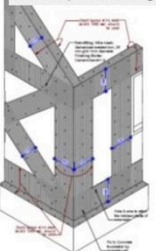
- ◆ Non-compliance construction can be appeared during inspection.
- ◆ Need of Correction / Retrofitting.
- ◆ Correction manual should be developed as a part of retrofitting.

➤ Developing Retrofitting Method

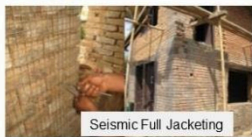


Wire mesh in Nepal

Seismic partial Jacking



Seismic Full Jacking



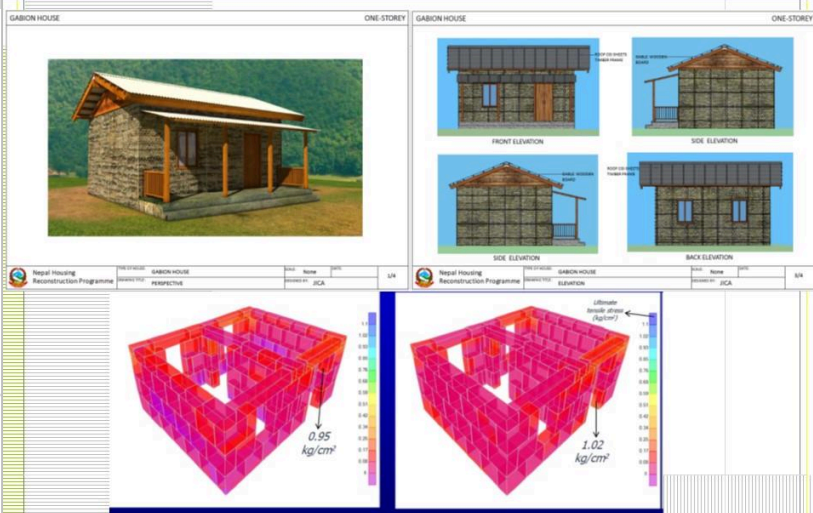
Seismic Full Jacking

- CGI mesh is widely available in Nepal.
- Unit price is around 350R p/m2 (1.6mm @ 20mm)
Various size is available.
- The application of best type of reinforcing, i.e., seismic partial reinforcing, seismic whole reinforcing, seismic band reinforcing based on the vulnerability of houses.



Shaking Table Test

Jacking (Full/Partial)



Jacking (Gabion type)



Pull Down Test